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Hippocampal Trilaminar Cells: Structure, Location, Connectivity, Function, and Markers

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Introduction

Hippocampal trilaminar cells are named for a three-layered hippocampal arrangement. In the material provided here, the term “trilaminar” directly refers to a triangular hippocampal region made up of lateral and medial layers with a triangular part between them. (1 source)

The term *trilaminar* in this context points to a specific three-part anatomical layout in the hippocampal formation rather than, by itself, giving a full definition of one neuron type. Meskenaite et al. describe a “triangular area” of the hippocampal formation that corresponds to the “trilaminar complex,” defined as lateral and medial layers with a triangular part between them, within the V-shaped structure of the medial wall. (Meskenaite et al., 2016) This provides the basic anatomical meaning of the term and sets up the later description of hippocampal trilaminar cells in terms of where they sit, how they are built, and how they connect within hippocampal circuits. (Meskenaite et al., 2016)

Evidence

(Meskenaite et al., 2016)

Age-Dependent Neurogenesis and Neuron Numbers within the Olfactory Bulb and Hippocampus of Homing Pigeons

> *"the triangular area of the hippocampus (HP tr), which we used for sampling in this study, was delineated according to the hippocampal boundaries of Karten Karten and Hodos (1967). (B) Triangular area of the hippocampal formation (HP tr) on two antero-posterior planes (Karten and Hodos, 1967), exact delineations can be found in (Atoji et al., 2004) and (Herold et al., 2015). and Hodos (1967), and corresponding to the trilaminar complex (lateral and medial layers with triangular part between them) of the hippocampal formation by (Atoji et al., 2004) within the V-shaped structure in the medial wall (Figure 1B), see also (Herold et al., 2015)."*

Identity, cell type, and anatomical location

*Hippocampal trilaminar cells are a specific **GABAergic interneuron/projection-neuron type**, not just any cell with axon branches in three layers. They are found mainly in **CA1**, with somata most often at the **oriens/alveus border** or in deep **stratum oriens**, and they also send long-range projections, especially to the **subiculum**. (8 sources)*

Trilaminar cells were defined as a **specific interneuron class** on the basis of soma orientation plus a characteristic laminar layout of dendrites and axon, and later authors argued that the term should be reserved for this cell type rather than used loosely for any neuron with axons in three layers. In the classic

CA1 description, the cell had **dendrites confined to stratum oriens** and **axon collaterals in stratum oriens, pyramidale, and radiatum**, which is the core feature behind the name "trilaminar." (Ferraguti et al., 2005) (Sik et al., 1995) They are part of the broader set of **long-range GABAergic hippocampal neurons** rather than purely local interneurons, because in addition to local CA1 axon branches they project out of CA1, most clearly to the **subiculum**. (Jinno et al., 2007) (Ferraguti et al., 2005)

Most reports place trilaminar cell bodies in **CA1 stratum oriens**, often close to the **alveus/oriens border**. Ferraguti et al. identified a strongly **M2-positive**, **mGluR8-decorated** interneuron in this position and identified it morphologically as a trilaminar cell. (Ferraguti et al., 2005) Jinno et al. likewise refer to trilaminar cells as a CA1 cell population in **stratum oriens** that projects to the subiculum. (Jinno et al., 2007) More recent work extends this picture by showing cells with trilaminar-like molecular features in both **CA1 and CA3** in rats and mice, though CA1 remains the best described location in the material provided here. (Katona et al., 2020)

There is also some evidence that the label has not always been used in exactly the same way across studies. A review cited here notes a **deep CA1 pyramidal-cell subpopulation** in rats and mice that had been termed "trilaminar cells," with large somata and a local axon plexus in stratum oriens. (Slomianka et al., 2011) By contrast, the interneuron literature in the other cited papers treats trilaminar cells as **GABAergic**, **M2-rich**, **somatostatin-negative** neurons, often with long-range projection to the subiculum. (Ferraguti et al., 2005) (Jinno et al., 2007) (Katona et al., 2020) Taken together, the strongest and most consistent identity in these sources is that hippocampal trilaminar cells are a **specialized long-range GABAergic cell type of deep CA1**, usually in **oriens/alveus**, with local axon spread across three CA1 layers and projection beyond CA1, especially to the subiculum. (Sik et al., 1995) (Ferraguti et al., 2005) (Jinno et al., 2007) (Wick et al., 2019)

A few newer and secondary sources add that **Chrm2/M2-positive** GABAergic neurons that fit the trilaminar profile are sparse and are found mainly in **SO and SP of CA1-CA3**, with some morphological variation across cells. (Li et al., 2026) Cells described as "radial trilaminar cells" have also been noted at low density in **CA1 stratum pyramidale**. (Kaliszewski et al., 2016) These points suggest some heterogeneity in exact soma position and shape, but they do not change the main identity: trilaminar cells are a rare, deep-layer hippocampal **GABAergic projection interneuron class** centered on **CA1 oriens/alveus**. (Katona et al., 2020)

Evidence

(Ferraguti et al., 2005)

Metabotropic Glutamate Receptor 8-Expressing Nerve Terminals Target Subsets of GABAergic Neurons in the Hippocampus

"Based on the orientation of the soma, the laminar distribution of the dendrites and the axon, this cell closely resembled a cell described previously in a similar position and named trilaminar cell (Sik et al., 1995). Although other properties cannot be directly compared in the two studies, we maintain this terminology here and propose that the name "trilaminar" is only applied to this specific cell type (Somogyi et al., 2005) and not to other different cells that have axons in three layers"

".The dendritic field covered ≈ 930m mediolaterally and 1500 m anteroposteriorly in the CA1 area and is among the largest fields in our sample of cells (Fig. 8j). The dendrites were irregularly and sparsely spiny, with most of the spines of the drumstick-shaped variety (Fig. 8c). In the CA1 area and subiculum, the axon spanned ≈ 3.3mm in the rostrocaudal direction. The axon was most dense at the level of the dendritic field and most widespread in the stratum oriens passing into the stratum pyramidale. In the vicinity of the cell, several major axon collaterals reached the border of the stratum radiatum and lacunosum moleculare, where they turned back and returned to the stratum oriens. Bouton-bearing collaterals were present in all innervated layers and densest in the stratum oriens (Fig. 8d,e)"

".A surprising feature of the cell was an extensive projection to the subiculum. Three main axon branches traveled caudally in the alveus, giving short branches infrequently to the stratum oriens and pyramidale and eventually branching extensively in all layers and in a large area of the subiculum, including the molecular layer. Another collateral proceeded in the alveus/fimbria to the most-rostral pole of the hippocampus and was lost in the stratum oriens; whether it left the hippocampus could not be ascertained"

".In vivo recording and labeling of an mGluR8-decorated and strongly M2-positive interneuron revealed a trilaminar cell with complex spike bursts during theta oscillations and strong discharge during sharp wave/ripple events. The trilaminar cell had a large projection from the CA1 area to the subiculum and a preferential innervation of interneurons in the CA1 area in addition to pyramidal cell somata and dendrites"

".The most prominent targets of mGluR8-positive synapses are the horizontal M2 receptor-immunoreactive neurons in the CA1 oriens/alveus and in the CA3 area. The mGluR8 decoration of these cells is both denser and stronger than that of other cell types. The visualization of the axon and dendrites of one of these cells made it possible to identify them as the trilaminar cells (Sik et al., 1995). The firing of complex spike bursts together with the axonal distribution, synapses onto pyramidal cell somata and onto a high proportion of interneurons, and their septal projection clearly separate this cell type from other interneurons (Somogyi et al., 2005). Trilaminar cells are the fourth independent identified source of GABA onto pyramidal cell somata (Somogyi et al., 2005). The preferential targeting of other interneurons by trilaminar cells, together with their highfrequency firing on sharp wave-associated ripples, points to the possibility that this cell might contribute to the specific silencing of axo-axonic or O-LM cells (Klausberger et al., 2003), which are inhibited during ripple episodes. Because O-LM cells are mostly in the alveus/oriens, whereas the trilaminar cell targets were in and around the pyramidal cell layer, axo-axonic cells are the most likely candidates"

".The source of excitatory drive to trilaminar cells is not known. During theta activity, the prolonged glutamate release from all hippocampal afferents would increase ambient glutamate levels and activate mGluR8, leading to suppression of glutamate release to trilaminar cells and hence contributing to their low firing rate. During slow wave sleep and related brain states, the episodic firing of pyramidal cells interspersed with long periods of low activity would only increase glutamate levels during ripples, allowing the trilaminar cells to be activated by glutamatergic afferents, followed by mGluR8-mediated suppression."

(Sik et al., 1995)

Hippocampal CA1 interneurons: an in vivo intracellular labeling study

"The trilaminar interneuron had axon collaterals in strata radiatum, oriens and pyramidale with its dendrites confined to stratum oriens. Commissural stimulation evoked an early EPSP-IPSP-late depolarizing potential sequence in this cell. All interneurons formed symmetric synapses with their targets at the electron microscopic level."

(Jinno et al., 2007)

Neuronal Diversity in GABAergic Long-Range Projections from the Hippocampus

"Cell T85a has been described previously as a mGluR8-decorated, M2expressing, SOM-negative trilaminar cell (Ferraguti et al., 2005). The dendrites of oriens-retrohippocampal cells extended horizontally in str. oriens and the local axonal arbors, exhibiting en passant boutons in the CA1 area, were distributed in the CA1 str. oriens and radiatum, to a lesser extent in str. pyramidale"

"(3) trilaminar cells in stratum oriens that project to the subiculum (Ferraguti et al., 2005);"

(Katona et al., 2020)

Synaptic organisation and behaviour-dependent activity of mGluR8a-innervated GABAergic trilaminar cells projecting from the hippocampus to the subiculum

"In the hippocampal CA1 area, the GABAergic trilaminar cells have their axon distributed locally in three layers and also innervate the subiculum. Trilaminar cells have a high level of somato-dendritic muscarinic M2 acetylcholine receptor, lack somatostatin expression and their presynaptic inputs are enriched in mGluR8a"

". Here we demonstrate that (1) GABAergic neurons with the molecular features of trilaminar cells are present in CA1 and CA3 in both rats and mice. (2) Trilaminar cells receive mGluR8a-enriched GABAergic inputs, e.g. from the medial septum, which are probably susceptible to hetero-synaptic modulation of neurotransmitter release by group III mGluRs. (3) An electron microscopic analysis identifies trilaminar cell output synapses with specialised postsynaptic densities and a strong bias towards interneurons as targets, including parvalbumin-expressing cells in the CA1 area. (4) Recordings in freely moving rats revealed the network state-dependent segregation of trilaminar cell activity, with reduced firing during movement, but substantial increase in activity with prolonged burst firing (> 200 Hz) during slow wave sleep"

". Amongst, the neurons described as long-range projecting and GABAergic, the strongest resemblance to trilaminar cells are the M2+ subgroup of VIP-LRP cells (Type I) of CGE origin identified in the mouse (Francavilla et al. 2018;Luo et al. 2019). These neurons, like the trilaminar cells, are SST immunonegative with their somata and horizontally oriented dendritic trees at the border between stratum oriens/ alveus and receive inputs from mGluR8a+ terminals. The local axon collaterals of these cells are concentrated within stratum oriens/alveus, with only a small proportion entering strata pyramidale and radiatum"

". What distinguishes trilaminar cells molecularly from M2+ VIP-LRP cells (Luo et al. 2019) is the lack of expression of VIP, neuropeptide Y and CB in the cell somata and the very high density of their mGluR8a+ inputs, as we quantified here"

". However, single cell mRNA sequencing in the mouse hippocampal CA1 has revealed an isolated cluster of CGE-derived neurons in stratum oriens lacking Sst or Pvalb and expressing Chrm2 and Ntng1, a molecular profile similar to that of trilaminar cells (Harris et al., 2017)(Qian et al., 2018)."

(Slomianka et al., 2011)

Hippocampal pyramidal cells: the reemergence of cortical lamination

"In area CA1, Sánchez-Aguilera et al. (2014) identified a deep subpopulation of pyramidal cells that expresses a broad range of markers, among them Ctip2, Sox5, muscarinic acetylcholine receptor M1, and several neuropeptides, including somatostatin and neuropeptide Y. In rats and mice, these cells were shown to have a large body and a local axon plexus in stratum oriens and were termed "trilaminar cells". In the monkey, they may correspond to "profusely branching or radiatum cells" (Buckmaster and Amaral 2001). Over 50 % of the trilaminar cells project to subcortical targets, including the septum, olfactory bulb, and hypothalamus (Sik et al. 1994, 1995), and they innervate local interneurons (Gulyás et al. 1998). They are thought to coordinate the septohippocampal networks during sleep-related oscillations (Sik et al. 1994)."

(Wick et al., 2019)

Novel long-range inhibitory nNOS-expressing hippocampal cells

"Other hippocampal GABAergic projection neurons, including trilaminar cells and oriens-retrohippocampal cells, are commonly associated with the expression of M2R (Jinno et al., 2007)(Klausberger, 2009)."

(Li et al., 2026)

Single-nucleus profiling reveals a core disease signature and cell type–specific vulnerabilities in early Rett syndrome

"Trilaminar neurons are long-range projection INs that span multiple hippocampal layers. As illustrated in the schematic (Fig. 6B), these cells have been previously described as located in the stratum oriens (SO) and stratum pyramidale (SP) and with their dendritic and axonal projections extending through stratum radiatum (SR) to reach the distal stratum lacunosum-moleculare (SLM) layer of the hippocampus, hence their designation as "trilaminar" (Booker et al., 2018). This strategic anatomical positioning places them at a key junction to modulate information flow across hippocampal layers"

".We confirmed that Chrm2 colocalizes with Gad2 and labels a sparse number of INs located mainly within the SO and SP layers of CA1-CA3 (Fig. 6, C to E). We identified a subpopulation of Chrm2 neurons that did not colocalize with Sst or Pvalb neurons and localized mostly in the CA1 subregion (Fig. 6, D and E)"

".We found two morphologically distinct GAD2 + /CHRM2 + populations that both colocalized with MeCP2: one located in the upper SO layers with horizontal projecting dendrites and another in or near the SP layers displaying perpendicular projecting dendrites that spanned the SO and SR hippocampal layers (Fig. 6G). This morphological heterogeneity within Chrm2 + trilaminar INs suggests potential functional diversity within the population"

".Chrm2 + INs were enriched for core synaptic and neuron function pathways (Fig. 6J)"

".The shared changes between NULL and HET NEG trilaminar INs predominantly affect genes involved in neurotransmitter release and synaptic function, with most being down-regulated (fig. S6, B to D)."

(Kaliszewski et al., 2016)

SOD1 Lysine 123 Acetylation in the Adult Central Nervous System

"Higher magnification of the hippocampus indicated Ac-K123 SOD1 localization to CA1 stratum pyramidale (SP) cell bodies and processes within the CA1 stratum radiatum (SR; Figure 4B). The CA1 SP mostly consists of pyramidal neurons, which receive input from Schaffer collaterals and send axons to the subiculum and entorhinal cortex. Also present within this layer, at a much lower density, are the somata and processes of local circuit inhibitory interneurons that include axo-axonic, basket, bistratified and radial trilaminar cells."

Structure and morphology

Trilaminar cells are deep hippocampal interneurons with soma and dendrites mainly in stratum oriens, while their axon spreads across three CA1 layers—oriens, pyramidale, and radiatum. They are usually elongated or oblong cells, often with horizontal dendrites, but some reports describe radial/perpendicular variants, suggesting modest morphological heterogeneity. (8 sources)

Morphologically, the defining feature of a trilaminar cell is the mismatch between a **restricted dendritic tree** and a **broad three-layer axonal arbor**. In the original CA1 description, the dendrites were confined to **stratum oriens**, whereas the axon gave collaterals in **strata oriens, pyramidale, and radiatum**, which is the core reason for the name “trilaminar.” (Sik et al., 1995) This basic pattern was later reaffirmed, with the caution that “trilaminar” should be reserved for this specific cell type rather than any neuron whose axon happens to enter three layers. (Ferraguti et al., 2005)

Their somata are typically described as **oblong or elongated** and positioned deep in CA1, often around the **oriens/alveus border**. In slice-based descriptions, trilaminar and related backprojecting interneurons in CA1

SO were noted to have **oblong cell bodies** with dendrites extending **perpendicular to pyramidal neurons**. (Hewitt et al., 2021) (Hewitt et al._1, 2021) Other anatomical descriptions emphasize that at the SO/alveus border these cells can look **bipolar** in one cutting plane and **multipolar** in another, with processes largely confined to a narrow deep band; near the CA1/subiculum border, some cell bodies shift upward toward the pyramidal layer. (Slomianka et al., 2013)

The **dendritic shape** shows some variation across reports. A well-filled CA1 trilaminar cell described by Ferraguti et al. had a very large dendritic field within CA1, spanning about **930 µm mediolaterally** and **1500 µm anteroposteriorly**, and its dendrites were **irregularly and sparsely spiny**, with many **drumstick-shaped spines**. (Ferraguti et al., 2005) Reviews summarizing the field note that trilaminar cells in stratum oriens can have either **horizontal dendrites** like the original Sik cell or more **radial dendrites** in other descriptions. (Booker et al., 2018) A more recent mouse study likewise reported two **Chrm2-positive** morphologies that fit the trilaminar class: one in upper SO with **horizontal-projecting dendrites**, and another in or near SP with **perpendicular dendrites** spanning SO and SR. (Li et al., 2026)

The **axon** is extensive locally and can also become long range. In Ferraguti et al., the CA1/subicular axon extended for **more than 3.3 mm rostrocaudally**, was **densest near the dendritic field**, and was especially widespread in **stratum oriens** as it passed into **stratum pyramidale**; near the cell, several major collaterals reached the **radiatum/lacunosum-moleculare border** and then turned back toward oriens. Bouton-bearing collaterals were present in all innervated layers but were **densest in stratum oriens**. (Ferraguti et al., 2005) A review summarizing quantitative reconstructions similarly described a **long, dense axon** distributed across oriens, pyramidale, and radiatum, with a substantial fraction of output in radiatum and a meaningful share of synapses in pyramidale. (Booker et al., 2018) (Sik et al., 1995) Francavilla et al. also note that the rat trilaminar cell had a **large soma**, **horizontally running dendrites at the O/A border**, and an axon running from **O/A through PYR to RAD**, with a strong bias toward **proximal radiatum**. (Francavilla et al., 2018)

At the ultrastructural level, trilaminar cells fit their GABAergic identity: like the other interneurons examined in the same study, they formed **symmetric synapses** onto their targets. (Sik et al., 1995) Overall, their structure is notable for combining a deep-layer, often elongated soma; dendrites largely centered in oriens; and a broad axon that links multiple CA1 layers and can extend onward into adjacent hippocampal output regions. (Sik et al., 1995) (Ferraguti et al., 2005) (Booker et al., 2018)

Evidence

(Sik et al., 1995)

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".The source of excitatory drive to trilaminar cells is not known. During theta activity, the prolonged glutamate release from all hippocampal afferents would increase ambient glutamate levels and activate mGluR8, leading to suppression of glutamate release to trilaminar cells and hence contributing to their low firing rate. During slow wave sleep and related brain states, the episodic firing of pyramidal cells interspersed with long periods of low activity would only increase glutamate levels during ripples, allowing the trilaminar cells to be activated by glutamatergic afferents, followed by mGluR8-mediated suppression."

(Hewitt et al., 2021)

High and low expression of the hyperpolarization activated current (Ih) in mouse CA1 stratum oriens interneurons

"Trilaminar and back propagating interneurons are also found in CA1 SO and present with oblong cells bodies with dendrites that extend perpendicular to pyramidal CA1 neurons (Sik et al., 1994)(Sik et al., 1995). It is unlikely that LTH cells are trilaminar cells considering trilaminar cells do not express a sag and rebound response (Gloveli et al., 2005)."

(Hewitt et al._1, 2021)

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The hippocampus of the eastern rock sengi: cytoarchitecture, markers of neuronal function, principal cell numbers, and adult neurogenesis

"the number of cells located at the so/alveus border increases in CA1 (Figures 5F,J). They often appear bipolar in sections cut perpendicular to the hippocampal long-axis but multipolar in sections cut parallel to the alveus (Figure 5I) and extend a network of processes that is confined to the narrow band defined by the cell bodies (Figures 5H,J). At the CA1/subiculum border, cell bodies ascend toward the CA1pcl (Figure 5L). Fine processes and a few cell bodies extend into the proximal extreme of the subiculum (Figure 5L). Generally, stained cells are very rare in the remaining layers of CA1 (Figure 5H). A fine plexus of fibers is seen in the deep part of CA1so (Figure 5J)."

(Booker et al., 2018)

Morphological diversity and connectivity of hippocampal interneurons

"With somata located in the str. oriens, trilaminar cells have either horizontal (Sik et al., 1995) or radial (Ali et al. 1998) dendrites (Fig. 2g). In the horizontal orientation, they resemble SOM INs. The axon of the str. oriens trilaminar cells resembles those described above (section BCCK trilaminar INs^), having a long (56,000 µm, 15,767 boutons) and dense axon within the str. oriens (12.8%), pyramidale (16.7%) and radiatum (68.4%); it should be noted that axons in the str. pyramidale formed a significant proportion of synaptic contacts in this layer (Sik et al., 1995), with up to 19% of synapses contacting PC somata (Ferraguti et al., 2005). Little is known of the connectivity of trilaminar neurons but they receive a strong excitatory Schaffer collateral input as well as strong inhibition (Sik et al., 1994)(Sik et al., 1995). In addition to the dense local axon, trilaminar cells may also produce long-range retrohippocampal connections to CA3 and dentate gyrus (DG) (see also BINs with local and long-range projecting axons^section), ramifying in the dendritic fields of principal cells (Sik et al., 1994). It remains, however, unknown if this is a characteristic of all trilaminar cells or of only a subset."

(Li et al., 2026)

Single-nucleus profiling reveals a core disease signature and cell type–specific vulnerabilities in early Rett syndrome

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> *".The shared changes between NULL and HET NEG trilaminar INs predominantly affect genes involved in neurotransmitter release and synaptic function, with most being down-regulated (fig. S6, B to D)."*

(Francavilla et al., 2018)

Connectivity and network state-dependent recruitment of long-range VIP-GABAergic neurons in the mouse hippocampus

> *"One candidate is the trilaminar cell identified previously in the rat hippocampus (Sik et al., 1995), which was shown to express M2R in the somato-dendritic membrane, was decorated with mGluR8a-containing terminals and projected to the subiculum (Ferraguti et al., 2005). This cell had a large soma with horizontally running dendrites at the O/A border and an axon innervating the CA1 from O/A through PYR to RAD. However, the distribution of the trilaminar cell axon in the rat was biased toward proximal RAD (70%; Sik et al., 1995)"*

Connectivity and synaptic targets

Trilaminar cells are inhibitory CA1 neurons with both local and long-range outputs. Locally they target many interneurons, but they also synapse onto pyramidal-cell somata and dendrites; beyond CA1, they send a strong projection especially to the subiculum. (7 sources)

A key feature of trilaminar cells is that they are not purely local interneurons. Their local axon branches span CA1 oriens, pyramidale, and radiatum, but they also send long-range axons out of CA1, most clearly to the subiculum. In the best-described example, the axon was dense near the home dendritic field, ran widely through oriens and pyramidale, sent collaterals toward radiatum/lacunosum-moleculare, and then gave an extensive caudal projection through the alveus that branched broadly across all layers of the subiculum, including the molecular layer. (Ferraguti et al., 2005) (Sik et al., 1995) Jinno et al. likewise summarize trilaminar cells in stratum oriens as CA1 neurons that project to the subiculum, and later reviews also treat subicular projection as a defining connection of the class. (Jinno et al., 2007) (Katona et al., 2020)

Their output is inhibitory at the synaptic level. Sik et al. reported that the trilaminar interneuron, like the other interneurons they studied, formed **symmetric synapses**, which is the ultrastructural sign expected for GABAergic output. (Sik et al., 1995) Within CA1, trilaminar cells do not appear to act mainly like classic basket cells with a near-exclusive pyramidal-cell focus. Instead, Ferraguti et al. found a **preferential innervation of interneurons** in CA1, while still making contacts onto **pyramidal-cell somata and dendrites**. They further note that trilaminar cells are an identified source of GABA onto pyramidal-cell somata, but with a strong bias toward interneuron targets. (Ferraguti et al., 2005) This interneuron bias was strengthened by later electron-microscopy work showing a strong preference for interneuron targets, including **parvalbumin-expressing cells** in CA1. (Katona et al., 2020)

This target choice suggests that trilaminar cells can shape CA1 activity both **directly** and **indirectly**. Directly, they inhibit pyramidal cells through somatic and dendritic contacts. Indirectly, by inhibiting other

interneurons, they can produce layer- and state-specific disinhibitory effects in the local network. Ferraguti et al. specifically suggested that their strong firing during ripple events, together with their interneuron preference, could help silence interneuron classes that are themselves suppressed during ripples, with axo-axonic cells raised as likely candidates. (Ferraguti et al., 2005)

On the input side, available evidence is still limited, but trilaminar cells clearly receive strong synaptic drive. In vivo commissural stimulation evoked an **early EPSP, then IPSP, then late depolarizing potential** in the original trilaminar cell, showing that it is embedded in both excitatory and inhibitory CA1 circuitry. (Sik et al., 1995) Reviews summarize this as strong **Schaffer collateral excitation** together with strong inhibition, although they also note that connectivity data remain sparse. (Booker et al., 2018) Ferraguti et al. argued that the exact excitatory afferent source was still unknown, but proposed that glutamatergic input and presynaptic mGluR8-dependent control could regulate when trilaminar cells are recruited across network states. (Ferraguti et al., 2005)

Finally, there is some evidence for added long-range complexity beyond the CA1–subiculum pathway. Some reviews discuss reports of projections to subcortical targets such as the septum, olfactory bulb, and hypothalamus, and possible retrohippocampal connections toward CA3 and dentate gyrus, but these links are less consistently established in the material here than the CA1/subiculum projection. (Slomianka et al., 2011) (Booker et al., 2018) (Francavilla et al., 2018) Overall, the most solid connectivity picture is of a **deep CA1 GABAergic projection interneuron** that spreads inhibition across several CA1 layers, strongly influences other interneurons, still contacts pyramidal cells, and extends a notable inhibitory output into the subiculum. (Ferraguti et al., 2005) (Jinno et al., 2007) (Katona et al., 2020)

Evidence

(Ferraguti et al., 2005)

Metabotropic Glutamate Receptor 8-Expressing Nerve Terminals Target Subsets of GABAergic Neurons in the Hippocampus

"Based on the orientation of the soma, the laminar distribution of the dendrites and the axon, this cell closely resembled a cell described previously in a similar position and named trilaminar cell (Sik et al., 1995). Although other properties cannot be directly compared in the two studies, we maintain this terminology here and propose that the name "trilaminar" is only applied to this specific cell type (Somogyi et al., 2005) and not to other different cells that have axons in three layers"

".The dendritic field covered ≈ 930m mediolaterally and 1500 m anteroposteriorly in the CA1 area and is among the largest fields in our sample of cells (Fig. 8j). The dendrites were irregularly and sparsely spiny, with most of the spines of the drumstick-shaped variety (Fig. 8c). In the CA1 area and subiculum, the axon spanned ≈ 3.3mm in the rostrocaudal direction. The axon was most dense at the level of the dendritic field and most widespread in the stratum oriens passing into the stratum pyramidale. In the vicinity of the cell, several major axon collaterals reached the border of the stratum radiatum and lacunosum moleculare, where they turned back and returned to the stratum oriens. Bouton-bearing collaterals were present in all innervated layers and densest in the stratum oriens (Fig. 8d,e)"

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".In vivo recording and labeling of an mGluR8-decorated and strongly M2-positive interneuron revealed a trilaminar cell with complex spike bursts during theta oscillations and strong discharge during sharp wave/ripple events. The trilaminar cell had a large projection from the CA1 area to the subiculum and a preferential innervation of interneurons in the CA1 area in addition to pyramidal cell somata and dendrites"

".The most prominent targets of mGluR8-positive synapses are the horizontal M2 receptor-immunoreactive neurons in the CA1 oriens/alveus and in the CA3 area. The mGluR8 decoration of these cells is both denser and stronger than that of other cell types. The visualization of the axon and dendrites of one of these cells made it possible to identify them as the trilaminar cells (Sik et al., 1995). The firing of complex spike bursts together with the axonal distribution, synapses onto pyramidal cell somata and onto a high proportion of interneurons, and their septal projection clearly separate this cell type from other interneurons (Somogyi et al., 2005). Trilaminar cells are the fourth independent identified source of GABA onto pyramidal cell somata (Somogyi et al., 2005). The preferential targeting of other interneurons by trilaminar cells, together with their highfrequency firing on sharp wave-associated ripples, points to the possibility that this cell might contribute to the specific silencing of axo-axonic or O-LM cells (Klausberger et al.,

2003), which are inhibited during ripple episodes. Because O-LM cells are mostly in the alveus/oriens, whereas the trilaminar cell targets were in and around the pyramidal cell layer, axo-axonic cells are the most likely candidates"

".The source of excitatory drive to trilaminar cells is not known. During theta activity, the prolonged glutamate release from all hippocampal afferents would increase ambient glutamate levels and activate mGluR8, leading to suppression of glutamate release to trilaminar cells and hence contributing to their low firing rate. During slow wave sleep and related brain states, the episodic firing of pyramidal cells interspersed with long periods of low activity would only increase glutamate levels during ripples, allowing the trilaminar cells to be activated by glutamatergic afferents, followed by mGluR8-mediated suppression."

(Sik et al., 1995)

Hippocampal CA1 interneurons: an in vivo intracellular labeling study

"The trilaminar interneuron had axon collaterals in strata radiatum, oriens and pyramidale with its dendrites confined to stratum oriens. Commissural stimulation evoked an early EPSP-IPSP-late depolarizing potential sequence in this cell. All interneurons formed symmetric synapses with their targets at the electron microscopic level."

(Jinno et al., 2007)

Neuronal Diversity in GABAergic Long-Range Projections from the Hippocampus

"Cell T85a has been described previously as a mGluR8-decorated, M2expressing, SOM-negative trilaminar cell (Ferraguti et al., 2005). The dendrites of oriens-retrohippocampal cells extended horizontally in str. oriens and the local axonal arbors, exhibiting en passant boutons in the CA1 area, were distributed in the CA1 str. oriens and radiatum, to a lesser extent in str. pyramidale"

"(3) trilaminar cells in stratum oriens that project to the subiculum (Ferraguti et al., 2005);"

(Katona et al., 2020)

Synaptic organisation and behaviour-dependent activity of mGluR8a-innervated GABAergic trilaminar cells projecting from the hippocampus to the subiculum

"In the hippocampal CA1 area, the GABAergic trilaminar cells have their axon distributed locally in three layers and also innervate the subiculum. Trilaminar cells have a high level of somato-dendritic muscarinic M2 acetylcholine receptor, lack somatostatin expression and their presynaptic inputs are enriched in mGluR8a"

". Here we demonstrate that (1) GABAergic neurons with the molecular features of trilaminar cells are present in CA1 and CA3 in both rats and mice. (2) Trilaminar cells receive mGluR8a-enriched GABAergic inputs, e.g. from the medial septum, which are probably susceptible to hetero-synaptic modulation of neurotransmitter release by group III mGluRs. (3) An electron microscopic analysis identifies trilaminar cell output synapses with specialised postsynaptic densities and a strong bias towards interneurons as targets, including parvalbumin-expressing cells in the CA1 area. (4) Recordings in freely moving rats revealed the network state-dependent segregation of trilaminar cell activity, with reduced firing during movement, but substantial increase in activity with prolonged burst firing (> 200 Hz) during slow wave sleep"

". Amongst, the neurons described as long-range projecting and GABAergic, the strongest resemblance to trilaminar cells are the M2+ subgroup of VIP-LRP cells (Type I) of CGE origin identified in the mouse (Francavilla et al. 2018;Luo et al. 2019). These neurons, like the trilaminar cells, are SST immunonegative with their somata and horizontally oriented dendritic trees at the border between stratum oriens/ alveus and receive inputs from mGluR8a+ terminals. The local axon collaterals of these cells are concentrated within stratum oriens/alveus, with only a small proportion entering strata pyramidale and radiatum"

". What distinguishes trilaminar cells molecularly from M2+ VIP-LRP cells (Luo et al. 2019) is the lack of expression of VIP, neuropeptide Y and CB in the cell somata and the very high density of their mGluR8a+ inputs, as we quantified here"

". However, single cell mRNA sequencing in the mouse hippocampal CA1 has revealed an isolated cluster of CGE-derived neurons in stratum oriens lacking Sst or Pvalb and expressing Chrm2 and Ntng1, a molecular profile similar to that of trilaminar cells (Harris et al., 2017)(Qian et al., 2018)."

(Booker et al., 2018)

Morphological diversity and connectivity of hippocampal interneurons

"With somata located in the str. oriens, trilaminar cells have either horizontal (Sik et al., 1995) or radial (Ali et al. 1998) dendrites (Fig. 2g). In the horizontal orientation, they resemble SOM INs. The axon of the str. oriens trilaminar cells resembles those described above (section BCCK trilaminar INs^), having a long (56,000 µm, 15,767 boutons) and dense axon within the str. oriens (12.8%), pyramidale (16.7%) and radiatum (68.4%); it should be noted that axons in the str. pyramidale formed a significant proportion of synaptic contacts in this layer (Sik et al., 1995), with up to 19% of synapses contacting PC somata (Ferraguti et al., 2005). Little is known of the connectivity of trilaminar neurons but they receive a strong excitatory Schaffer collateral input as well as strong inhibition (Sik et al., 1994)(Sik et al., 1995). In addition to the dense local axon, trilaminar cells may also produce long-range retrohippocampal connections to CA3 and dentate gyrus (DG) (see also BINs with local and long-range projecting axons^section),

ramifying in the dendritic fields of principal cells (Sik et al., 1994). It remains, however, unknown if this is a characteristic of all trilaminar cells or of only a subset."

(Slomianka et al., 2011)

Hippocampal pyramidal cells: the reemergence of cortical lamination

"In area CA1, Sánchez-Aguilera et al. (2014) identified a deep subpopulation of pyramidal cells that expresses a broad range of markers, among them Ctip2, Sox5, muscarinic acetylcholine receptor M1, and several neuropeptides, including somatostatin and neuropeptide Y. In rats and mice, these cells were shown to have a large body and a local axon plexus in stratum oriens and were termed "trilaminar cells". In the monkey, they may correspond to "profusely branching or radiatum cells" (Buckmaster and Amaral 2001). Over 50 % of the trilaminar cells project to subcortical targets, including the septum, olfactory bulb, and hypothalamus (Sik et al. 1994, 1995), and they innervate local interneurons (Gulyás et al. 1998). They are thought to coordinate the septohippocampal networks during sleep-related oscillations (Sik et al. 1994)."

(Francavilla et al., 2018)

Connectivity and network state-dependent recruitment of long-range VIP-GABAergic neurons in the mouse hippocampus

"One candidate is the trilaminar cell identified previously in the rat hippocampus (Sik et al., 1995), which was shown to express M2R in the somato-dendritic membrane, was decorated with mGluR8a-containing terminals and projected to the subiculum (Ferraguti et al., 2005). This cell had a large soma with horizontally running dendrites at the O/A border and an axon innervating the CA1 from O/A through PYR to RAD. However, the distribution of the trilaminar cell axon in the rat was biased toward proximal RAD (70%; Sik et al., 1995)"

Physiological activity and function

Trilaminar cells are state-dependent inhibitory neurons. They tend to fire relatively little during movement/theta but become strongly active, often in bursts, during slow-wave sleep and sharp wave/ripple events, where they likely help reshape CA1 and subicular inhibition. (7 sources)

Physiologically, trilaminar cells are notable for **network-state-dependent firing** rather than uniform activity across behavior. In vivo, an identified CA1 trilaminar cell showed **complex spike bursts during theta oscillations** but **strong discharge during sharp wave/ripple events** (Ferraguti et al., 2005). More recent recordings in freely moving rats similarly found **reduced firing during movement** and a **substantial increase in activity during slow-wave sleep**, including **prolonged burst firing above 200 Hz** (Katona et al., 2020). Together, these findings place trilaminar cells among hippocampal interneurons that are especially recruited during offline or sleep-related population events rather than during active locomotor theta states (Ferraguti et al., 2005) (Katona et al., 2020).

Their synaptic responses also show that they sit at a point where excitation and inhibition converge. In the original in vivo study, commissural stimulation evoked an **early EPSP, then an IPSP, then a late depolarizing potential**, indicating that trilaminar cells are driven by both excitatory and inhibitory pathways within CA1 circuitry (Sik et al., 1995). Reviews based on those data summarize them as receiving **strong Schaffer collateral excitation together with strong inhibition**, though the exact afferent sources remain incompletely defined (Booker et al., 2018) (Sik et al., 1995).

Functionally, their activity pattern and target choice suggest that they can help **reconfigure inhibition during sharp wave/ripple episodes**. Because trilaminar cells fire strongly during ripples and preferentially innervate other interneurons, Ferraguti et al. proposed that they may help suppress interneuron classes that are themselves silent during ripples, especially **axo-axonic cells** and possibly **O-LM cells**, thereby changing when pyramidal cells are inhibited or disinhibited across layers (Ferraguti et al., 2005). Katona et al. extend this picture by showing that their output has a **strong bias toward interneurons, including parvalbumin-expressing cells**, which supports a role in indirect control of CA1 ensemble timing through inhibition of other inhibitory neurons (Katona et al., 2020).

Their molecular physiology suggests one possible mechanism for this state dependence. Ferraguti et al. proposed that during theta, prolonged glutamate release could raise ambient glutamate enough to activate **presynaptic mGluR8**, suppressing excitatory drive to trilaminar cells and helping keep their firing low; during slow-wave sleep, by contrast, more episodic glutamatergic activity during ripples could recruit them

transiently before mGluR8-mediated suppression develops (Ferraguti et al., 2005). Consistent with this idea, Katona et al. report that trilaminar cells receive **mGluR8a-enriched inputs**, including GABAergic inputs from the medial septum, indicating that both local and long-range modulatory systems can tune when these cells are active (Katona et al., 2020).

Because trilaminar cells also project beyond CA1, especially to the **subiculum**, their firing is well placed to couple local hippocampal population states to downstream targets (Ferraguti et al., 2005) (Katona et al., 2020). Secondary sources therefore suggest that they may help **coordinate septohippocampal and hippocampo-subicular networks during sleep-related oscillations** (Slomianka et al., 2011). More generally, they fit the broader class of long-range GABAergic projection neurons that can transmit rhythm-linked inhibitory timing between connected regions (Katona et al., 2020) (Katona et al., 2017).

Older slice work on related “radial trilaminar” cells also suggested a possible role in **contrasting Schaffer-collateral-driven versus entorhinal-driven CA1 states**, though this was a more tentative functional idea and not specific to the classic CA1 oriens trilaminar cell definition (Hajos et al., 1997). Overall, the strongest evidence supports a function in **state-dependent inhibitory control**, especially during **sharp wave/ripple and slow-wave-sleep activity**, where trilaminar cells can influence both local CA1 microcircuits and downstream hippocampal output regions (Ferraguti et al., 2005) (Katona et al., 2020) (Sik et al., 1995).

Evidence

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(Katona et al., 2017)

Behavior-dependent activity patterns of GABAergic long-range projecting neurons in the rat hippocampus

> *"Long-range glutamatergic and GABAergic projections participate in temporal coordination of neuronal activity in distributed cortical areas. In the hippocampus, GABAergic neurons project to the medial septum and retrohippocampal areas. Many GABAergic projection cells express somatostatin (SOM+) and, together with locally terminating SOM+ bistratified and O-LM cells, contribute to dendritic inhibition of pyramidal cells. We tested the hypothesis that diversity in SOM+ cells reflects temporal specialization during behavior using extracellular single cell recording and juxtacellular neurobiotin-labeling in freely moving rats. We have demonstrated that rare GABAergic projection neurons discharge rhythmically and are remarkably diverse. During sharp wave-ripples, most projection cells, including a novel SOM+ GABAergic back-projecting cell, increased their activity similar to bistratified cells, but unlike O-LM cells. During movement, most projection cells discharged along the descending slope of theta cycles, but some fired at the trough jointly with bistratified and O-LM cells. The specialization of hippocampal SOM+ projection neurons complements the action of local interneurons in differentially phasing inputs from the CA3 area to CA1 pyramidal cell dendrites during sleep and wakefulness. Our observations suggest that GABAergic projection cells mediate the behavior- and network state-dependent binding of neuronal assemblies amongst functionally-related brain regions by transmitting local rhythmic entrainment of neurons in CA1 to neuronal populations in other areas. © 2016 The Authors Hippocampus Published by Wiley Periodicals, Inc."*

(Hajos et al., 1997)

Synaptic Communication among Hippocampal Interneurons: Properties of Spontaneous IPSCs in Morphologically Identified Cells

> *"Interneurons with appearance similar to our str. radiatum and radial trilaminar cells have been previously alluded to, but with less complete anatomical details (Kawaguchi andHama, 1987, 1988;(Bergles et al., 1996)"*
>
> *".The multipolar cells with axon clouds in strata radiatum and lacunosum-moleculare have not been described previously, although cells with stellate-like appearance containing neuropeptide Y (NPY) and immunoreactive for substance P receptor (SPR) have been reported without any axon staining (Acsády et al., 1997). The similar dendritic morphology may mean that these multipolar cells labeled in the present study belong to the NPY and SPR-immunopositive category of interneurons"*
>
> *".Their function may be to enhance the contrast between a Schaffer collateral-driven state and the entorhinal-driven state of CA1 pyramidal cells."*

Molecular markers and distinguishing features

*The most useful marker profile for hippocampal trilaminar cells is **strong M2/Chrm2 expression**, **dense mGluR8a-positive input decoration**, and **lack of somatostatin**. They are also set apart by being **GABAergic long-range projection interneurons** with a three-layer local axon pattern, strong interneuron targeting, and some physiological traits that help separate them from nearby oriens cell types. (11 sources)*

- **Core positive marker: muscarinic M2 receptor / Chrm2**
- Trilaminar cells are strongly associated with **somato-dendritic M2 receptor expression** and are often described as **strongly M2-positive**. This is one of the clearest markers used to recognize the class in CA1. (Ferraguti et al., 2005) (Jinno et al., 2007) (Katona et al., 2020)
- Later work also links trilaminar-like cells to **Chrm2-positive GABAergic neurons** in CA1-CA3, especially in **SO and SP**, supporting M2/Chrm2 as a practical molecular handle for this population. (Li et al., 2026)
- **Core negative marker: somatostatin (SST/SOM)**
- A repeated distinguishing feature is that trilaminar cells are **somatostatin-negative**. This helps separate them from several other oriens long-range projection interneurons that are SOM-positive. (Jinno et al., 2007) (Katona et al., 2020)
- Ferraguti et al. also noted that among M2-positive oriens neurons, the subgroup strongly decorated by mGluR8 terminals was mostly distinct from the **somatostatin-expressing minority** that was not similarly innervated, helping define the trilaminar phenotype more narrowly. (Ferraguti et al., 2005)
- **Characteristic input marker: dense mGluR8a-positive bouton decoration**
- A notable identifying feature is the very strong **mGluR8 / mGluR8a-enriched presynaptic input** onto trilaminar cells. Ferraguti et al. describe these neurons as among the most prominent **mGluR8-decorated**

M2-positive cells in CA1 oriens/alveus, and the morphologically identified recorded cell was both **strongly M2-positive** and **mGluR8-decorated**. (Ferraguti et al., 2005)

- Katona et al. confirm that trilaminar cells receive **mGluR8a-enriched GABAergic inputs**, including from the **medial septum**, and argue that the **very high density** of these mGluR8a-positive inputs helps distinguish them from related M2-positive long-range cells. (Katona et al., 2020)
- **GABAergic identity**
- Trilaminar cells belong to the hippocampal **GABAergic projection interneuron** group rather than to pyramidal-cell classes. This is supported by their placement among long-range interneurons and by work showing **Chrm2 colocalization with Gad2** in sparse SO/SP interneurons that fit the trilaminar profile. (Sik et al., 1995) (Li et al., 2026)
- **Likely transcriptomic signature**
- Single-cell transcriptomic work cited by Katona et al. points to a CA1 oriens cluster that lacks **Sst** and **Pvalb** while expressing **Chrm2** and **Ntng1**, which appears consistent with the trilaminar-cell profile. (Katona et al., 2020) (Harris et al., 2017) (Qian et al., 2018)
- **Markers usually absent compared with nearby interneuron classes**
- Trilaminar cells are reported as lacking **somatostatin**, and newer work suggests they also differ from some related M2-positive long-range projection cells by lacking **VIP**, **neuropeptide Y**, and **calbindin (CB)** in the soma. (Jinno et al., 2007) (Katona et al., 2020)
- They also appear distinct from a **Pvalb-positive** identity; transcriptomic discussion places them in a **Sst-/Pvalb-** space rather than among classic parvalbumin basket or axo-axonic populations. (Katona et al., 2020) (Harris et al., 2017)
- **How to tell them apart from related M2-positive projection cells**
- Some M2-positive **VIP long-range projection neurons** resemble trilaminar cells in soma position, horizontal dendrites, SST-negativity, and mGluR8a-positive inputs. What best separates classic trilaminar cells, according to Katona et al., is the **lack of VIP/NPY/CB expression** together with a **very dense mGluR8a input pattern** and the classic trilaminar axon layout. (Katona et al., 2020) (Francavilla et al., 2018)
- **How to tell them apart from other oriens interneurons by combined features**
- Ferraguti et al. argued that the combination of **complex spike bursting**, **three-layer local axon distribution**, **projection to the subiculum**, **somatic output to pyramidal cells**, and **high interneuron targeting** clearly separates trilaminar cells from other interneurons. (Ferraguti et al., 2005) (Sik et al., 1995)
- Reviews of hippocampal GABAergic projection neurons also note that **M2R expression** is commonly associated with trilaminar cells, reinforcing its value as a distinguishing feature when used with morphology and connectivity. (Wick et al., 2019) (Jinno et al., 2007)
- **Useful physiological discriminator**
- A practical extra feature is that trilaminar cells reportedly **do not show a sag and rebound response**, which can help distinguish them from some other CA1 SO interneurons in slice studies. (Hewitt et al., 2021) (Hewitt et al._1, 2021)
- **Bottom-line identification rule**
- In the papers provided here, the safest way to identify a trilaminar cell is not with one marker alone, but with a **marker bundle**: **GABAergic**, **strong M2/Chrm2-positive**, **SOM-negative**, **heavily decorated by mGluR8a-positive inputs**, plus the classic **oriens-centered dendrites and axon in oriens, pyramidale, and**

radiatum, often with **projection to subiculum**. (Ferraguti et al., 2005) (Jinno et al., 2007) (Katona et al., 2020) (Sik et al., 1995)

Evidence

(Ferraguti et al., 2005)

Metabotropic Glutamate Receptor 8-Expressing Nerve Terminals Target Subsets of GABAergic Neurons in the Hippocampus

"Based on the orientation of the soma, the laminar distribution of the dendrites and the axon, this cell closely resembled a cell described previously in a similar position and named trilaminar cell (Sik et al., 1995). Although other properties cannot be directly compared in the two studies, we maintain this terminology here and propose that the name "trilaminar" is only applied to this specific cell type (Somogyi et al., 2005) and not to other different cells that have axons in three layers"

".The dendritic field covered ≈ 930m mediolaterally and 1500 m anteroposteriorly in the CA1 area and is among the largest fields in our sample of cells (Fig. 8j). The dendrites were irregularly and sparsely spiny, with most of the spines of the drumstick-shaped variety (Fig. 8c). In the CA1 area and subiculum, the axon spanned ≈ 3.3mm in the rostrocaudal direction. The axon was most dense at the level of the dendritic field and most widespread in the stratum oriens passing into the stratum pyramidale. In the vicinity of the cell, several major axon collaterals reached the border of the stratum radiatum and lacunosum moleculare, where they turned back and returned to the stratum oriens. Bouton-bearing collaterals were present in all innervated layers and densest in the stratum oriens (Fig. 8d,e)"

".A surprising feature of the cell was an extensive projection to the subiculum. Three main axon branches traveled caudally in the alveus, giving short branches infrequently to the stratum oriens and pyramidale and eventually branching extensively in all layers and in a large area of the subiculum, including the molecular layer. Another collateral proceeded in the alveus/fimbria to the most-rostral pole of the hippocampus and was lost in the stratum oriens; whether it left the hippocampus could not be ascertained"

".In vivo recording and labeling of an mGluR8-decorated and strongly M2-positive interneuron revealed a trilaminar cell with complex spike bursts during theta oscillations and strong discharge during sharp wave/ripple events. The trilaminar cell had a large projection from the CA1 area to the subiculum and a preferential innervation of interneurons in the CA1 area in addition to pyramidal cell somata and dendrites"

".The most prominent targets of mGluR8-positive synapses are the horizontal M2 receptor-immunoreactive neurons in the CA1 oriens/alveus and in the CA3 area. The mGluR8 decoration of these cells is both denser and stronger than that of other cell types. The visualization of the axon and dendrites of one of these cells made it possible to identify them as the trilaminar cells (Sik et al., 1995). The firing of complex spike bursts together with the axonal distribution, synapses onto pyramidal cell somata and onto a high proportion of interneurons, and their septal projection clearly separate this cell type from other interneurons (Somogyi et al., 2005). Trilaminar cells are the fourth independent identified source of GABA onto pyramidal cell somata (Somogyi et al., 2005). The preferential targeting of other interneurons by trilaminar cells, together with their highfrequency firing on sharp wave-associated ripples, points to the possibility that this cell might contribute to the specific silencing of axo-axonic or O-LM cells (Klausberger et al., 2003), which are inhibited during ripple episodes. Because O-LM cells are mostly in the alveus/oriens, whereas the trilaminar cell targets were in and around the pyramidal cell layer, axo-axonic cells are the most likely candidates"

".The source of excitatory drive to trilaminar cells is not known. During theta activity, the prolonged glutamate release from all hippocampal afferents would increase ambient glutamate levels and activate mGluR8, leading to suppression of glutamate release to trilaminar cells and hence contributing to their low firing rate. During slow wave sleep and related brain states, the episodic firing of pyramidal cells interspersed with long periods of low activity would only increase glutamate levels during ripples, allowing the trilaminar cells to be activated by glutamatergic afferents, followed by mGluR8-mediated suppression."

(Jinno et al., 2007)

Neuronal Diversity in GABAergic Long-Range Projections from the Hippocampus

"Cell T85a has been described previously as a mGluR8-decorated, M2expressing, SOM-negative trilaminar cell (Ferraguti et al., 2005). The dendrites of oriens-retrohippocampal cells extended horizontally in str. oriens and the local axonal arbors, exhibiting en passant boutons in the CA1 area, were distributed in the CA1 str. oriens and radiatum, to a lesser extent in str. pyramidale"

"(3) trilaminar cells in stratum oriens that project to the subiculum (Ferraguti et al., 2005);"

(Katona et al., 2020)

Synaptic organisation and behaviour-dependent activity of mGluR8a-innervated GABAergic trilaminar cells projecting from the hippocampus to the subiculum

"In the hippocampal CA1 area, the GABAergic trilaminar cells have their axon distributed locally in three layers and also innervate the subiculum. Trilaminar cells have a high level of somato-dendritic muscarinic M2 acetylcholine receptor, lack somatostatin

expression and their presynaptic inputs are enriched in mGluR8a"

". Here we demonstrate that (1) GABAergic neurons with the molecular features of trilaminar cells are present in CA1 and CA3 in both rats and mice. (2) Trilaminar cells receive mGluR8a-enriched GABAergic inputs, e.g. from the medial septum, which are probably susceptible to hetero-synaptic modulation of neurotransmitter release by group III mGluRs. (3) An electron microscopic analysis identifies trilaminar cell output synapses with specialised postsynaptic densities and a strong bias towards interneurons as targets, including parvalbumin-expressing cells in the CA1 area. (4) Recordings in freely moving rats revealed the network state-dependent segregation of trilaminar cell activity, with reduced firing during movement, but substantial increase in activity with prolonged burst firing (> 200 Hz) during slow wave sleep"

". Amongst, the neurons described as long-range projecting and GABAergic, the strongest resemblance to trilaminar cells are the M2+ subgroup of VIP-LRP cells (Type I) of CGE origin identified in the mouse (Francavilla et al. 2018;Luo et al. 2019). These neurons, like the trilaminar cells, are SST immunonegative with their somata and horizontally oriented dendritic trees at the border between stratum oriens/ alveus and receive inputs from mGluR8a+ terminals. The local axon collaterals of these cells are concentrated within stratum oriens/alveus, with only a small proportion entering strata pyramidale and radiatum"

". What distinguishes trilaminar cells molecularly from M2+ VIP-LRP cells (Luo et al. 2019) is the lack of expression of VIP, neuropeptide Y and CB in the cell somata and the very high density of their mGluR8a+ inputs, as we quantified here"

". However, single cell mRNA sequencing in the mouse hippocampal CA1 has revealed an isolated cluster of CGE-derived neurons in stratum oriens lacking Sst or Pvalb and expressing Chrm2 and Ntng1, a molecular profile similar to that of trilaminar cells (Harris et al., 2017)(Qian et al., 2018)."

(Li et al., 2026)

Single-nucleus profiling reveals a core disease signature and cell type–specific vulnerabilities in early Rett syndrome

"Trilaminar neurons are long-range projection INs that span multiple hippocampal layers. As illustrated in the schematic (Fig. 6B), these cells have been previously described as located in the stratum oriens (SO) and stratum pyramidale (SP) and with their dendritic and axonal projections extending through stratum radiatum (SR) to reach the distal stratum lacunosum-moleculare (SLM) layer of the hippocampus, hence their designation as "trilaminar" (Booker et al., 2018). This strategic anatomical positioning places them at a key junction to modulate information flow across hippocampal layers"

".We confirmed that Chrm2 colocalizes with Gad2 and labels a sparse number of INs located mainly within the SO and SP layers of CA1-CA3 (Fig. 6, C to E). We identified a subpopulation of Chrm2 neurons that did not colocalize with Sst or Pvalb neurons and localized mostly in the CA1 subregion (Fig. 6, D and E)"

".We found two morphologically distinct GAD2 + /CHRM2 + populations that both colocalized with MeCP2: one located in the upper SO layers with horizontal projecting dendrites and another in or near the SP layers displaying perpendicular projecting dendrites that spanned the SO and SR hippocampal layers (Fig. 6G). This morphological heterogeneity within Chrm2 + trilaminar INs suggests potential functional diversity within the population"

".Chrm2 + INs were enriched for core synaptic and neuron function pathways (Fig. 6J)"

".The shared changes between NULL and HET NEG trilaminar INs predominantly affect genes involved in neurotransmitter release and synaptic function, with most being down-regulated (fig. S6, B to D)."

(Sik et al., 1995)

Hippocampal CA1 interneurons: an in vivo intracellular labeling study

"The trilaminar interneuron had axon collaterals in strata radiatum, oriens and pyramidale with its dendrites confined to stratum oriens. Commissural stimulation evoked an early EPSP-IPSP-late depolarizing potential sequence in this cell. All interneurons formed symmetric synapses with their targets at the electron microscopic level."

(Harris et al., 2017)

Classes and continua of hippocampal CA1 inhibitory neurons revealed by single-cell transcriptomics

"Understanding any brain circuit will require a categorization of its constituent neurons. In hippocampal area CA1, at least 23 classes of GABAergic neuron have been proposed to date. However, this list may be incomplete; additionally, it is unclear whether discrete classes are sufficient to describe the diversity of cortical inhibitory neurons, or whether continuous modes of variability are also required. We studied the transcriptomes of 3663 CA1 inhibitory cells, revealing 10 major GABAergic groups that divided into 49 fine-scale clusters. All previously described and several novel cell classes were identified, with three previously-described classes unexpectedly found to be identical. A common continuous mode of variation modulated the expression of many genes, reflecting a range from metabolically highly active faster-spiking cells that proximally target pyramidal cells, to slower-spiking cells that target distal dendrites or interneurons. Within classes, cells showed further variability along the continuum and this correlated with their spiking phenotype. Surprisingly, several cell types previously described as discrete classes represented ranges along this continuum. These results elucidate the complexity of inhibitory neurons in one of the simplest cortical structures, and show that

characterizing these cells requires continuous modes of variation as well as discrete cell classes."

(Qian et al., 2018)

A spatial atlas of inhibitory cell types in mouse hippocampus

"Understanding the function of a tissue requires knowing the spatial organization of its constituent cell types. In the cerebral cortex, single-cell RNA sequencing (scRNA-seq) has revealed the genome-wide expression patterns that define its many, closely related cell types, but cannot reveal their spatial arrangement. Here we introduce probabilistic cell typing by in situ sequencing (pciSeq), an approach that leverages prior scRNA-seq classification to identify cell types using multiplexed in situ RNA detection. We applied this method to map the inhibitory neurons of hippocampal area CA1, a cell system critical for memory function, for which ground truth is available from extensive prior work identifying the laminar organization of subtly differing cell types. Our method confidently identified 16 interneuron classes, in a spatial arrangement closely matching ground truth. This method will allow identifying the spatial organization of fine cell types across the brain and other tissues."

(Francavilla et al., 2018)

Connectivity and network state-dependent recruitment of long-range VIP-GABAergic neurons in the mouse hippocampus

"One candidate is the trilaminar cell identified previously in the rat hippocampus (Sik et al., 1995), which was shown to express M2R in the somato-dendritic membrane, was decorated with mGluR8a-containing terminals and projected to the subiculum (Ferraguti et al., 2005). This cell had a large soma with horizontally running dendrites at the O/A border and an axon innervating the CA1 from O/A through PYR to RAD. However, the distribution of the trilaminar cell axon in the rat was biased toward proximal RAD (70%; Sik et al., 1995)"

(Wick et al., 2019)

Novel long-range inhibitory nNOS-expressing hippocampal cells

"Other hippocampal GABAergic projection neurons, including trilaminar cells and oriens-retrohippocampal cells, are commonly associated with the expression of M2R (Jinno et al., 2007)(Klausberger, 2009)."

(Hewitt et al., 2021)

High and low expression of the hyperpolarization activated current (Ih) in mouse CA1 stratum oriens interneurons

"Trilaminar and back propagating interneurons are also found in CA1 SO and present with oblong cells bodies with dendrites that extend perpendicular to pyramidal CA1 neurons (Sik et al., 1994)(Sik et al., 1995). It is unlikely that LTH cells are trilaminar cells considering trilaminar cells do not express a sag and rebound response (Gloveli et al., 2005)."

(Hewitt et al._1, 2021)

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